

A pure state detects a nonzero square

MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self

theorem

Detects a nonzero element through its positive square without an algebra separability assumption.

Statement

Let A be a nonzero unital complex C^* -algebra, and let $a \in A$ be nonzero. There is a pure state φ on A such that $\varphi(a^*a) \neq 0$.

Here a state is a continuous complex-linear functional positive on positive elements and satisfying $\varphi(1) = 1$; pure means extreme in the real convex state space.

Assumptions

No separability assumption is made on A . The element detected in the conclusion is a^*a , rather than an assertion about $\varphi(a)$.

Conclusion

One functional φ is simultaneously a state, pure, and nonzero on a^*a .

Proof route

Choose a nonzero spectral value of the positive square, realize it as a character value on its generated commutative algebra, and extend that character to a pure state.

▼ Proof steps

1. Put $b = a^*a$. The C^* -identity gives $\|b\| = \|a\|^2 > 0$, so $b \neq 0$. It is self-adjoint. Its spectral radius equals $\|b\|$, and compactness of its spectrum supplies $z \in \sigma(b)$ with $|z| = \|b\| > 0$.
2. Let $D = C^*(1, b) \subseteq A$, the norm-closed unital $*$ -subalgebra generated by b . Since b is self-adjoint, D is commutative. The character-space identification with $\sigma(b)$ gives a character χ of D with $\chi(b) = z \neq 0$. These are the two applications in Spectral radius and the generated-algebra character map: self-adjointness supplies the norm-radius equality, and the chosen spectral point is an input to the surjective character map.
3. Apply Pure-state extension of a character to this closed commutative unital D and this character χ . It produces a state φ on A which is pure and satisfies $\varphi(x) = \chi(x)$ for every $x \in D$. Substituting the same $b = a^*a \in D$ gives

$$\varphi(a^*a) = \varphi(b) = \chi(b) = z \neq 0.$$

The first two outputs of the extension theorem give the remaining state and purity conditions.

Main citations

- Exact theorem and proof
- Pure-state extension of a character

Lean source signature (exact)

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theorem exists_pureState_nonzero_on_star_mul_self [Nontrivial A]
  {a : A} (ha : a ≠ 0) :
  ∃ phi : A →L[ℂ] ℂ,
    phi ∈ stateSpace A ∧ IsPureState A phi ∧ phi (star a * a) ≠ 0
```

In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
{a : A} (ha : a ≠ 0)	The specified nonzero element a .
∃ phi : A →L[ℂ] ℂ	One continuous complex-linear functional $\varphi : A \rightarrow \mathbb{C}$ is chosen.
phi ∈ stateSpace A	That functional is positive and normalized by $\varphi(1) = 1$.
IsPureState A phi	The same state is an extreme point for real convex combinations of states.
phi (star a * a) ≠ 0	Its value on the particular positive element a^*a is nonzero; this is the full final condition, not a claim about $\varphi(a)$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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